Releasing Telenomus podisi against the brown stink bug

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Table of contents

1 Overview 1

2 The model 2

3 A season without wasps 2

4 Release strategies 5

5 When parasitised eggs give new wasps 6

6 Deviations from the source 7

1 Overview

The neotropical brown stink bug, *Euschistus heros*, is the main pest of soybean in Brazil. The tiny wasp *Telenomus podisi* lays its eggs in stink bug eggs, and its larvae kill them. It is bred and released in fields as a biological control. Weber et al. (2022) built a lattice model of a soybean field in which individual stink bugs and wasps develop, move, lay eggs and die. They used it to compare 128 release strategies: how many releases, how many wasps, and in strips or at points how far apart. They published the model's C code with their preprint.

This implementation is a reproduction. One detail needs care. Read literally, the posted code never lets a stink bug stay in its cell, and an arriving insect overwrites one already there. That collapses the population to a few hundred adults, which cannot have produced the published results. Moving as the paper describes reproduces them, as shown below.

```
ENV["GKSwstype"] = "100"
using EcologicalCellularAutomata
using CellularAutomataCore
using Plots
using Statistics
```

2 The model

The field is 200 × 200 cells of 1 m², about 30 soybean plants each (4 ha). Only females are modelled. Each cell holds stink bug nymphs and adults as counts by age, up to 60 egg masses, and wasps as counts by age. Cells have room for 500 nymphs, 300 adults and 550 wasps. Each day:

- Within every cell (the model's rule), insects age and die: adult stink bugs at 3% a day, nymphs at 1.4%, wasps at 5.66%. Nymphs become adults at 30 days, adults die at 37 and wasps at 26. Egg masses hatch at 5 days, giving 3 female nymphs from 11 eggs. Parasitised eggs are removed when the wasps would emerge, at 14 days: in the source's main experiment the wasps' offspring are not followed.
- Stink bugs move up to 3 m along each axis, most staying put. Adults of at least 13 days lay a mass of 11 eggs nearby with probability 0.7.
- Each wasp flies to a random cell within 8 m that holds unparasitised eggs, or anywhere within 8 m if none does. From its second day it parasitises `L13.6 e^(-0.102 (age - 1))⌋` eggs there and in the cells around it, 12 on the second day falling to 1 on the 25th.
- 8000 adult stink bugs colonise the central 81 × 81 m over the first 10 days. Wasps are released in the central hectare from day 4, a week apart.

Movement, egg laying and parasitism happen insect by insect, and each depends on what the insects before it did. So they, colonisation and releases are a callback stored with the problem.

```
prob = problem(:telenomus_release; seed=1)
keys(prob.u0), keys(prob.kwargs)
```

```
((:nymphs, :adults, :masses, :wasps), (:callback,))
```

The state at time `t` is the field at the start of day `t`, so the source's step `s` is time `s + 1` here.

3 A season without wasps

We step an integrator through the season, recording the totals and the source's measure of damage each day. The measure is stink bugs of at least 0.5 cm (adults and nymphs older than 12 days) per m² in the central hectare, where 2 is the economic threshold. Three runs:

```
day_row(u) = merge(telenomus_counts(u), (; et=telenomus_et(u)))
size_class(u) = [sum(Int, u.adults[I]) + sum(Int, u.nymphs[I][13:end])
                 for I in CartesianIndices(u.adults)]
function season(prob; maps=())
    integ = init(prob; save_everystep=false, save_start=false)
    rows = [day_row(integ.u)]
    snapshots = Dict{Int,Matrix{Int}}()
    while integ.t < 80
        step!(integ)
        push!(rows, day_row(integ.u))
```

```
        integ.t in maps && (snapshots[integ.t] = size_class(integ.u))
    end
    return rows, snapshots
end
control = [season(problem(:telenomus_release; seed=s); maps=(31, 61, 80)) for s in
1:3]
step_of(i) = i - 2   # row i is time i - 1, the end of source step i - 2
peak(r, k) = (n=maximum(getfield.(r, k)), step=step_of(argmax(getfield.(r, k))))
[(final=round(r[end].et; digits=2), first_over=step_of(findfirst(x -> x.et > 2, r)),
  eggs=peak(r, :eggs), nymphs=peak(r, :nymphs), adults=peak(r, :adults))
 for r in first.(control)]
```

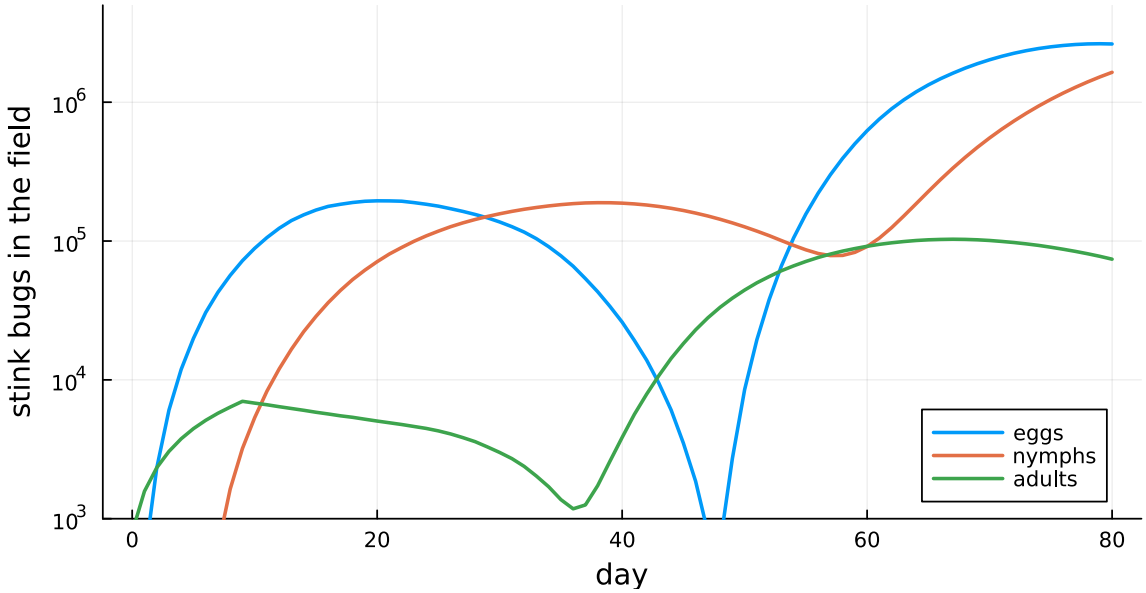
```
3-element Vector{@NamedTuple{final::Float64, first_over::Int64,
eggs::@NamedTuple{n::Int64, step::Int64}, nymphs::@NamedTuple{n::Int64,
step::Int64}, adults::@NamedTuple{n::Int64, step::Int64}}}:
 (final = 39.87, first_over = 25, eggs = (n = 2636788, step = 78), nymphs = (n =
 1641751, step = 79), adults = (n = 103071, step = 66))
 (final = 39.4, first_over = 25, eggs = (n = 2572548, step = 78), nymphs = (n =
 1608310, step = 79), adults = (n = 100799, step = 66))
 (final = 39.08, first_over = 25, eggs = (n = 2608001, step = 78), nymphs = (n =
 1618154, step = 79), adults = (n = 101749, step = 66))
```

Each row gives the final measure, the step at which it first exceeds the threshold, and the peak of each stage with its step. The source reports, over 15 runs:

- a final measure of 39.37 (± 0.13), about 18 times the threshold;
- the threshold first exceeded at step 26, one step after it is here;
- peaks of 2 597 177 eggs at step 78, 1 618 391 nymphs at step 79 and 101 526 adults at step 66.

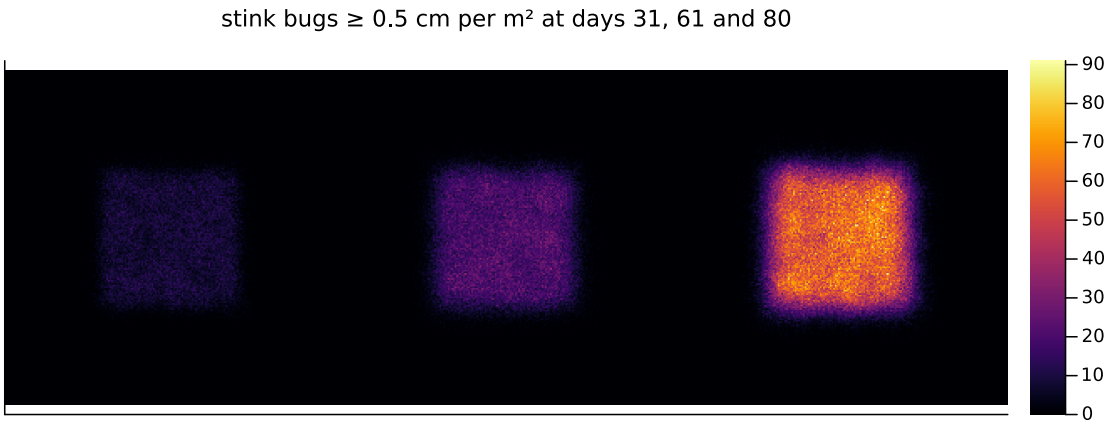
The runs here agree to within about 1.5%, at the same steps.

```
rows = first(control[1])
plot(0:80, [getfield.(rows, :eggs) getfield.(rows, :nymphs) getfield.(rows,
:adults)];
     label=["eggs" "nymphs" "adults"], lw=2, yscale=:log10, ylims=(1e3, 5e6),
     xlabel="day", ylabel="stink bugs in the field", legend=:bottomright,
     size=(650, 340))
```



Colonists lay eggs from the start. Their offspring take 35 days to become adults, so adults dwindle until the first generation emerges from about day 37. Its eggs then rise steeply, while the field holds no eggs for a few days around day 47, between the generations.

```
maps = control[1][2]
heatmap(hcat([maps[d] for d in (31, 61, 80)]...); aspect_ratio=1, c=:inferno,
        xticks=false, yticks=false, size=(750, 290), titlefontsize=10,
        title="stink bugs ≥ 0.5 cm per m² at days 31, 61 and 80")
```



Stink bugs stay close to the central square where they colonised the field: moving about half a metre a day, they spread only a few metres in a season, while their numbers there rise.

4 Release strategies

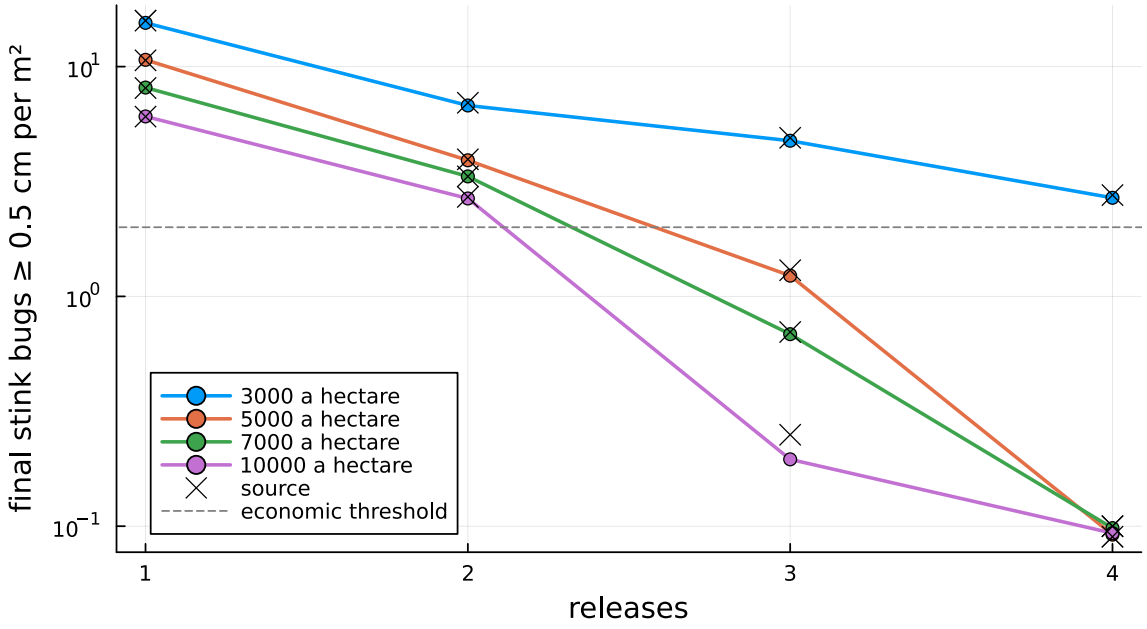
The source's Table S1 gives the final measure of each strategy. Here are the 16 strategies of strips 25 m apart: 1 to 4 releases of 3000 to 10 000 wasps a hectare, three runs each, in parallel:

```
final_et(prob) = telenomus_et(solve(prob; save_everystep=false,
save_start=false).u[end])
function strategy(; seeds=1:3, kwargs...)
    tasks = [Threads.@spawn final_et(problem(:telenomus_release; seed=s, kwargs...))
             for s in seeds]
    return fetch.(tasks)
end
sizes = (3000, 5000, 7000, 10000)
grid = [mean(strategy(; releases=r, per_ha=N)) for r in 1:4, N in sizes]
source = [15.85 10.65 8.07 6.06; 6.96 3.95 3.33 2.71; 4.90 1.30 0.70 0.25;
          2.76 0.10 0.09 0.10]
[(releases=r, per_ha=N, here=round(grid[r, k]; digits=2), source=source[r, k])
 for r in 1:4 for (k, N) in enumerate(sizes)]
```

```
16-element Vector{@NamedTuple{releases::Int64, per_ha::Int64, here::Float64,
source::Float64}}:
 (releases = 1, per_ha = 3000, here = 15.5, source = 15.85)
 (releases = 1, per_ha = 5000, here = 10.7, source = 10.65)
 (releases = 1, per_ha = 7000, here = 8.11, source = 8.07)
 (releases = 1, per_ha = 10000, here = 6.07, source = 6.06)
 (releases = 2, per_ha = 3000, here = 6.77, source = 6.96)
 (releases = 2, per_ha = 5000, here = 3.91, source = 3.95)
 (releases = 2, per_ha = 7000, here = 3.33, source = 3.33)
 (releases = 2, per_ha = 10000, here = 2.67, source = 2.71)
 (releases = 3, per_ha = 3000, here = 4.76, source = 4.9)
 (releases = 3, per_ha = 5000, here = 1.23, source = 1.3)
 (releases = 3, per_ha = 7000, here = 0.69, source = 0.7)
 (releases = 3, per_ha = 10000, here = 0.2, source = 0.25)
 (releases = 4, per_ha = 3000, here = 2.69, source = 2.76)
 (releases = 4, per_ha = 5000, here = 0.09, source = 0.1)
 (releases = 4, per_ha = 7000, here = 0.1, source = 0.09)
 (releases = 4, per_ha = 10000, here = 0.09, source = 0.1)
```

Every strategy agrees with the source to within about 0.2. As the source concludes, only three or four releases of at least 5000 wasps keep the final measure below the threshold of 2. Two large releases ($2 \times 10\,000$) do worse than three smaller ones (3×5000), even with more wasps in total: each release protects only the eggs laid while its wasps live.

```
plot(1:4, grid; marker=:circle, lw=2, label=permutedims(["$N a hectare" for N in
sizes]),
     xlabel="releases", ylabel="final stink bugs ≥ 0.5 cm per m²", yscale=:log10,
     xticks=1:4, size=(650, 360), legend=:bottomleft)
scatter!(vec(repeat(1:4, 1, 4)), vec(source); marker=:x, ms=6, color=:black,
         label="source")
hline!([2]; color=:gray, ls=:dash, label="economic threshold")
```



How far apart the releases are matters only when they are sparse. With four releases of 5000, the source gives 0.12 for points 25 m apart, 1.99 for points 50 m apart, and 0.87 for strips 50 m apart. Five runs each:

```
[(mode=m, spacing=s,
  runs=round.(strategy(; seeds=1:5, releases=4, per_ha=5000, mode=m, spacing=s);
              digits=2))
 for (m, s) in ((:points, 25), (:points, 50), (:strips, 50))]
```

```
3-element Vector{@NamedTuple{mode::Symbol, spacing::Int64, runs::Vector{Float64}}}:
 (mode = :points, spacing = 25, runs = [0.13, 0.12, 0.09, 0.13, 0.11])
 (mode = :points, spacing = 50, runs = [2.41, 2.39, 2.2, 2.55, 2.05])
 (mode = :strips, spacing = 50, runs = [0.84, 0.65, 0.61, 0.62, 0.76])
```

Points 25 m apart agree with the source. At 50 m the runs here differ from it by more than its standard errors (0.11 at most): points end near 2.3 against its 1.99, and strips near 0.7 against its 0.87. The ranking is the same, with sparse points the worst. With four points a hectare, much of the field lies beyond the wasps' 8 m reach, and each point gets more wasps than its cell holds, so they spill into the neighbouring cells. The source's results do not show why these two strategies differ here.

5 When parasitised eggs give new wasps

In the source's second configuration, parasitised eggs give a new generation of wasps (70% female, 82% surviving), which goes on searching. The source reports that every strategy then ends below the threshold, and that a single release of 7000 wasps, or two of 3000, may be enough. Three runs each:

```
[(releases=r, per_ha=N,
  final=round(mean(strategy(; releases=r, per_ha=N, offspring=true)); digits=2))
 for (r, N) in ((1, 3000), (1, 7000), (2, 3000))]
```

```
3-element Vector{@NamedTuple{releases::Int64, per_ha::Int64, final::Float64}}:
 (releases = 1, per_ha = 3000, final = 1.06)
 (releases = 1, per_ha = 7000, final = 0.4)
 (releases = 2, per_ha = 3000, final = 0.21)
```

Every strategy here also ends below the threshold, a single release of 3000 at about half of it.

6 Deviations from the source

```
foreach(d -> println("- ", d), model_info(:telenomus_release).deviations)
```

```
- Stink bugs move as the source describes them: a draw of no movement keeps the
insect in its cell, and a full cell makes it draw again. Read literally, the posted
C code never lets a stink bug stay and makes an arrival overwrite insects at the
destination; that collapses the population (to under 1000 adults) and cannot have
produced the published results.
- Insects of the same age in a cell are counts rather than individual storage
places, so a cell's insects move in order of age rather than of storage place; egg
masses keep their 60 places.
- Cells holding wasps are visited in a uniformly random order; the source draws a
random entry and searches outwards for the nearest unvisited one.
- Random integers are drawn uniformly; the source's integer draws, which scale
rand() and assume RAND_MAX = 32767, can rarely give one more than their range.
- Wasp fecundity is 13.62737 exp(-0.10221 (age - 1)), as in the code; the preprint
prints the product as a sum.
- The adult stink bug death rate is the code's 0.03 a day; the preprint's table (70%
viability over 24 days) implies 0.0147.
- The state at time t is the field at the start of day t, after that day's colonists
and releases, so the source's step s is time s + 1, and the colonists of the next
day are included in the first nine steps.
- As in the source, the economic-threshold measure counts 101 x 101 cells and
divides by 100 x 100.
```

Weber, Igor Daniel, Adriano Gomes Garcia, Adeney de Freitas Bueno, Regiane Cristina de Oliveira de Freitas Bueno, and Wesley Augusto Conde Godoy. 2022. “Release Strategies of *Telenomus Podisi* for Control of *Euschistus Heros*: A Computational Modeling Approach.” *Pest Management Science* 78 (11): 4544–56. https://doi.org/10.1002/ps.7074.